

David Dongjun Yang

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EDUCATION

Emory University

Bachelor of Science in Computer Science; GPA: 3.87/4.00

Aug 2022 – May 2026

Atlanta, GA

TECHNICAL SKILLS

Languages: Python, SQL, Java, C++, JavaScript, TypeScript, R, Bash

Focus: ML Engineer, ML Research Engineer, AI Engineer, Machine Learning, Deep Learning, Generative AI, LLMs, RAG

Methods: Computer Vision, Reinforcement Learning, Foundation Models, Self-Supervised Learning, Graph Neural Networks

Tools / Infra: PyTorch, TensorFlow, scikit-learn, LangChain, FAISS, Neo4j, SFT, LoRA, PEFT, GRPO, MLOps, AWS, Azure AI Foundry, GCP, Docker, Linux, Git, SQL

EXPERIENCE

Machine Learning Research Engineer

June 2026 – Current

Woven Solutions

Mclean, VA

- **Post-trained** the Flux2 Klein 9B image generative model with **distillation** and **reinforcement learning**, enhancing text-to-image, image-to-image generation capabilities.
- **Post-trained** Muse Glimmer 30B Vision Language model with **reinforcement learning** to enhance image and object detection task.

AI Engineer

Jan 2026 – June 2026

Emory University Office of Admission

Atlanta, GA

- Architected a multi-layer agentic conversation system in **Azure AI Foundry** leveraging **GPT 5.4**, **Azure SQL**, **MCP**, and text-embedding-3-large; integrated 70k+ rows of admissions data.
- Deployed a Microsoft Teams bot for counselors and marketing staff, achieving **92% human-rated task success** across 700+ production queries focused on market analysis and admission strategies.

Machine Learning Engineer Intern

June 2025 – Aug 2025

Naval Information Warfare Center (U.S. DOD)

Guam, USA

- Developed an **on-device RAG** pipeline for secure, localized question answering over 1.2k classified documents by adapting LoRA fine-tuned **Gemma3 4B** and **Llama3 8B** models.
- Engineered a **hybrid retrieval pipeline** utilizing **LangChain**, combining **FAISS** chunk semantic search with **Neo4j** graph entity linking to maximize evidence coverage and grounding.
- Evaluated and deployed the system to NIWC Engineers via Ollama, achieving **85% top-k retrieval relevance** with a **7.3s end-to-end local latency** on a 200+ query benchmark.

Research Assistant

Jan 2023 – Oct 2026

UCLA & Emory University

Atlanta, GA

- **Pre-trained** an **85M**-parameter foundation model for human sensor signals utilizing **ViT** encoders and **DINO/MAE** self-supervised learning, significantly improving missing-channel inference and out-of-domain transfer.
- Designed a harmonization pipeline across 9 public **multi-modular** PSG datasets, standardizing a 12-channel montage over **166.5K hours**, **21,482 recordings**, and **20M 30-second epochs**.
- Achieved **97.3 AUROC** and **90.4 AUPRC** on sleep staging under linear probing, improving to **97.9 / 92.0** under full **fine-tuning**, alongside performance gains on apnea and hypopnea detection.
- **Post-trained** an MAE-based **10M**-parameter infant EEG **foundation model** with continuous SSL, yielding **R2=0.91** and **12.7 MSE** for precise motor-score regression.

Machine Learning Research Intern

May 2024 – Aug 2024

UNC Greensboro @ GraLNA (U.S. NSF REU)

Greensboro, NC

- Developed an fMRI **image classification Graph Neural Network** architecture for neurodegenerative disease, utilizing a novel co-embedding aggregation layer for joint node/edge connectivity.
- Extracted representations from **459 patient scans** (ADNI/PPMI databases), generating node correlation matrices and **edge functional connectivity** mappings across **116 brain regions**.
- Benchmarked the model against CNN, GCN, CRGNN, and MGNN architectures, achieving state-of-the-art results including **84.4% precision** and **76.6% F1** on ADNI under 10-fold cross-validation.

PUBLICATIONS & PRESENTATIONS

- EEG Foundation Modeling for Early Prediction of Infant Motor Development** 2026
D. Yang and R. Xiao
8th Annual Pediatric Early Career Researcher Conference, Atlanta, GA
- On Pre-training and Scaling of Sleep Foundation Models** 2026
Z. Shuai, Z. Xu, D. Yang, W. Wang, and Y. Yang
Proceedings of the International Conference on Machine Learning (ICML)
- ConStruct: Structural Distillation of Foundation Models for Prototype-Based Weakly Supervised Histopathology Segmentation** 2026
K. Le, H. Thach, A. M. Vu, T. T. K. Vo, H. H. Huynh, D. Yang, M. H. N. Le, T. H. Nguyen, et al.
Proceedings of Medical Imaging with Deep Learning (MIDL)
- IMACT-CXR: An Interactive Multi-Agent Conversational Tutoring System for Chest X-Ray Interpretation** 2026
T. A. Le, A. M. Vu, D. Yang, A. Awasthi, and H. Van Nguyen
Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)
- Edge-Boosted Graph Learning for Functional Brain Connectivity Analysis** 2025
D. Yang, M. Abdelmegeed, J. Modl, and M. Kim
Proceedings of the IEEE International Symposium on Biomedical Imaging (ISBI)

PREPRINTS & UNDER REVIEW

- Infant-Adapted EEG Foundation Modeling for Developmental Status Classification in Infants Born Preterm** 2026
D. Yang, B. A. Smith, and R. Xiao
Under review — NeurIPS 2026 BrainBodyFM
- DualProtoSeg: Simple and Efficient Design with Text- and Image-Guided Prototype Learning for Weakly Supervised Histopathology Image Segmentation** 2025
A. M. Vu, K. P. Le, T. T. K. Vo, H. Thach, H. H. Nguyen, D. Yang, H. H. Huynh, et al.
arXiv preprint — arXiv:2512.10314
- Beyond the First Read: AI-Assisted Perceptual Error Detection in Chest Radiography Accounting for Interobserver Variability** 2025
A. Vutukuri, A. Awasthi, D. Yang, C. C. Wu, and H. Van Nguyen
arXiv preprint — arXiv:2506.13049

AWARDS

- Clinton Joiner Career Development Award—Best Research Rapid-Fire and Poster Presentation** 2026
8th Annual Pediatric Early Career Researcher Conference
Awarded for “EEG Foundation Modeling for Early Prediction of Infant Motor Development,” Arthur M. Blank Hospital, Children’s Healthcare of Atlanta, Atlanta, GA.
- NSF/CRA REU Travel Grant** 2025
Computing Research Association (CRA)